

Practice Problems

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x}{1+|x|}$. Let $g: [-1, 1] \rightarrow [-1, 1]$ be defined by $g(x) = \sin\left(\frac{\pi x}{2}\right)$. Consider the composite function $h: [-1, 1] \rightarrow \mathbb{R}$ defined by $h(x) = f(g(x))$. Which of the following statements is true about h ? (A) h is injective but not surjective. (B) h is surjective but not injective. (C) h is both injective and surjective. (D) h is neither injective nor surjective.

Let $f: \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$ be a function satisfying the functional equation $f(x) + f\left(\frac{1}{1-x}\right) = \frac{1}{x}$ for all $x \in \mathbb{R} \setminus \{0, 1\}$. Find the explicit expression for $f(x)$.

Let $f(x) = [x]$ (the greatest integer less than or equal to x) and $g(x) = |x|$. Consider the function $h(x) = f(g(x)) + g(f(x))$ for $x \in [-2, 2]$. Find the range of $h(x)$.

Let $A = \{1, 2, 3, 4, 5\}$. How many functions $f: A \rightarrow A$ are there such that $f(x) \neq x$ for exactly two values of $x \in A$?

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